**Transitioning from IPv4 to IPv6**

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When preparing a company network to transition from IPv4 to IPv6 there are many protocols and methods that need to be considered, researched, and documented. From ensuring supportability to setting up devices for traffic analytics there are various moving parts that need to be planned.

Without any devices in place it is impossible to send a request from an IPv4 address to an IPv6 address because they are naturally incompatible. There are 3 technologies to remedy this, Dual Stack Routers, Tunneling, and NAT Protocol Translation. A company can successfully transition if it implements these technologies appropriately and effectively. A dual stack router has its interface attached with both IPv4 and IPv6 address configured for use, allowing transition from IPv4 to IPv6 is possible. This gives all hosts a way to communicate with the sever without having to change their IP addresses. Tunneling is another way to communicate the transit network with different IP versions. NAT Protocol Translation is a technique that allows IPv4 and IPv6 networks to communicate via NAT-PT device. This device removes the header of the sending IP address and adds the receiver IP address so the Receiving version IP address can understand the request sent and vice versa. If the company applies one or any of these techniques properly to their network they will be able to transition smoothly and with efficient support.

To ensure that all network applications and TCP/IP services will support IPv6 there are different components that need to be checked and planned. First, check manufacturer's documentation for IPv6 readiness that all hardware and software that is being used can be upgraded to IPv6. Devices that need to be considered include the following: routers, firewalls, servers, and switches. Next, prepare network services such as sendmail, HTTP, DNS and LDAP for IPv6 support. Severs are considered IPv6 hosts and automatically configured by the Neighbor

Discovery protocol but it will be wise to swap out NICs that are due for a replacement. Then, verify network services that were ported to IPv6 and plan for tunnels within the network topology that needs to be supported by the IPv6 implementation.

When coordinating with other organizations and their networks it is important to communicate what protocols and devices they utilize and establish an agreement regarding traffic and security policies. There are many standards that should be upheld, companies needs ensure that outside organizations are upholding their end of those standard or communicate otherwise.

A new addressing approach while using the existing IPv4 topology is to create a IPv6 numbering scheme by mapping IPv4 subnets into equivalent IPv6 subnets. To save time, having stateless auto-configuration of the IPv6 addresses for interfaces is a great method using the Neighbor Discovery generation on the routers. Use sequential numbering scheme for interface IDs for servers, avoid regularly renumbering IPv4 network. To handle networking hardware compatibility, ensure all equipment connects network to everything and supports IPv6. In order to accomplish this, plan out what needs to a patch, firmware upgrade, or needs a hardware/software replacement.

There are many different tools available to monitor IPv6 traffic if a compatibility issue occurs. These monitors are also great in helping analyze and report IP network problems, have early detection of intrusion and research the behavior of third parties. It is important to select a tool that serves multiple purpose for financial efficiency.  Netflow 9, is a great traffic accounting protocol that supports many applications for usage billing, traffic analysts and capacity planning. IPFlow is a collector for the Netflow 9, it displays the flow statistics and logs flow data to disk, data aggregation and more.

Transitional methods from IPv4 to IPv6 are not perfect and will require some need to troubleshoot. But if a company takes the time to carefully investigate the documentation of hardware and software and consider its current limitations they will be able to plan successfully what needs to be done in order to slowly transition to IPv6. There are great technologies available to help the process such as NAT Protocol Translation or Dual Stack Routing, even with these technologies companies need to plan and implement upgrades or changes where needed so that there are a limited number of glitches.

## Reference

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